EXISTENCE AND STRUCTURE OF NASH EQUILIBRIA FOR SUPERMODULAR GAMES

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Abstract. Two theorems announced by Topkis about the topological description of sublattices are proved. They are applied to extend some classical results concerning the existence and the order structure of Nash equilibria of certain supermodular games, with some problems in Zhou's proof corrected.

1. Introduction

The purpose of this note is to generalize the classical results concerning the existence and the structure of Nash equilibria in supermodular games. Let $G = (N, \{S_i\}_{i\in N}, S, \{f_i\}_{i\in N})$ be a supermodular game (see Definition 3.8), where N is the set of players, S_i (resp. f_i) is the strategy set (resp. payoff function) of player i, and $S \subset \prod_{i\in N} S_i$ is the set of feasible strategy profiles. Importantly, we do not require necessarily $S = \prod_{i\in N} S_i$. Such games are sometimes called generalized games in the literature (see, e.g., [BK23, p.469]), and are very useful in economics (e.g., in general equilibrium). In particular, this class encompasses situations where the strategy choices of some player are constrained by the choices of other players.

A famous result of Topkis [Top79, Theorem 3.1] proved that the set of Nash equilibria of a supermodular game is nonempty and contains a maximum and a minimum, with many applications in game theory. Later, Zhou [Zho94, Theorem 2] proved that the set of Nash equilibria of such games is in fact a complete lattice, generalizing another result of Vives [Viv90, Theorem 4.1]. In Section 1, the player set N is assumed to be finite, as it is assumed in Facts 1.1 and 1.2, but we will relax this assumption in Theorems 4.11 and 4.12.

Fact 1.1 (Zhou)**.** *Assume that the game is in product form, i.e., $S = \prod_i S_i$. Assume that for each $i \in N$, S_i is compact for a topology τ_i finer than the interval topology, and the payoff function $f_i(\cdot, x_{-i}) : S_i \to \mathbb{R}$ is*

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upper semicontinuous relative to τ_i for each $x_{-i} \in \prod_{j \neq i} S_j$. Then the set of Nash equilibria is a nonempty complete lattice.

Based on Zhou's work, Topkis [Top98, Theorem 4.2.1] enhanced his previous result as follows.

Fact 1.2 (Topkis)**.** *Assume that each S_i is a sublattice of some Euclidean space $\mathbb{R}^{m_i}$ endowed with the subspace topology, and the set S of feasible joint strategies is nonempty and compact in the product topology on $\prod_{i \in N} S_i$. Assume that for each $x_{-i} \in S_{-i}$ and each $i \in N$, the payoff function $f_i(\cdot, x_{-i}) : \{x_i \in S_i : (x_i, x_{-i}) \in S\} \to \mathbb{R}$ is upper semicontinuous relative to the subspace topology. Then the set of Nash equilibria is a nonempty complete lattice. In particular, a greatest and a least equilibrium point exist.*

Hypotheses in Facts 1.1 and 1.2 are at the same time topological and order theoretical. The existence of equilibria for supermodular games (satisfying certain purely order-theoretical assumptions) is given in Veinott's unpublished work [Vei92, Ch. 10, Thm. 2]. As far as we are aware, Veinott's result does not contain Fact 1.1 or Fact 1.2.

Fact 1.2 restricts its conclusion to games with strategies in Euclidean spaces because the proof uses [Top98, Theorem 2.3.1]. By contrast, Fact 1.1 removes this restriction but is valid only for games in product form. Moreover, as pointed out in Section 4, the proof in [Zho94] is in fact insufficient for the stated generality. Here we generalize Facts 1.1 and 1.2 by allowing more flexible strategy sets.

Main Result (Theorem 4.2)**.** *For every $i \in N$, let τ_i be a topology on S_i finer than the interval topology. Assume that the sublattice $S \subset \prod_i S_i$ is closed and compact in the product topology of the τ_i. If for every $i \in N$, every $x \in S$, the function $f_i(\cdot, x_{-i}) : S_i(x_{-i}) \to \mathbb{R}$ is upper semicontinuous, then the set of Nash equilibria is a nonempty complete lattice.*

The paper is organized as follows: In Section 2, we give two theorems in Lattice theory, useful for our main result.[1] In Section 3, we recall standard definitions related to supermodular games. In Section 4, we prove our main theorem, as a consequence of Theorem 2.13. For games with infinitely many players, two results (Theorems 4.11 and 4.12) are also proved.

[1]These two theorems were stated by Topkis without published proofs, thus for the sake of completeness, we provide proofs.

2. Partial order and lattice

In this section, we prove two results stated by Topkis [Top98, p.31]. They make our main result possible.

Throughout this note, a set with a partial order is called a poset. The products of posets are always endowed with the product relation. We recall the following standard definitions:

Definition 2.1 (Lattice)**.** [Top98, p.13] A poset L is called a lattice if for every $x, y \in L$, both $\sup_L\{x, y\}$ and $\inf_L\{x, y\}$ exist in L, classically denoted by $x \vee_L y$ and $x \wedge_L y$ respectively.

Definition 2.2 (Complete lattice)**.** [Top98, p.29] A poset L is called a complete lattice if every nonempty subset A of L, both $\sup_L(A)$ and $\inf_L(A)$ exist. (Such an L is necessarily a lattice.)

Definition 2.3 (Increasing difference)**.** [Top98, p.42] Suppose that X and T are two posets. Let S be a subset of $X \times T$ and $f : S \to \mathbb{R}$ be a function. If for every $x \lneq x' \in X$, every $t \lneq t' \in T$ with $\{x, x'\} \times \{t, t'\} \subset S$ we have

$$f(x', t) + f(x, t') \leq f(x, t) + f(x', t'),$$

then we say f has increasing differences relative to $X \times T$.

Definition 2.4 (Interval topology)**.** [Top98, p.29] Let X be a poset. The topology on X generated by the subbasis for closed subsets comprised of closed intervals $(-\infty, x]$ and $[x, +\infty)$ (where x runs over X) is called the interval topology of X.

Definition 2.5 (Increasing correspondence)**.** [MS94, p.159] Let T (resp. L) be a poset (resp. lattice). A correspondence $\phi : T \twoheadrightarrow L$ is called increasing if for any $t \leq t'$ in T, every $x \in \phi(t)$ and every $x' \in \phi(t')$, one has

$$x \wedge x' \in \phi(t), \quad x \vee x' \in \phi(t').$$

Definition 2.6. [Top98, p.13, p.29] Let P be a poset, $S \subset P$ be a subset. If for each nonempty finite (resp. nonempty but possibly infinite) subset $A \subset S$, both $\sup_P(A)$ and $\inf_P(A)$ exist and belong to S, then S is called a sublattice (resp. subcomplete sublattice[2]) of P.

Fact 1.1 has been proved by Zhou using lattice theoretic Statement 2.7, which is wrong.

Statement 2.7.

[2]and also called *closed in the lattice theoretical sense* in [Zho94, p.296]

(1) ([Zho94, Remark 2]) Any sublattice of a complete lattice is closed in the lattice theoretical sense if and only if it is topologically closed in the interval topology.
(2) (the second paragraph of the proof of [Zho94, Thm. 2]) In the interval topology of a complete lattice, a compact subset is closed.

A counterexample to both points in Statement 2.7 is Example 2.8.

Example 2.8. Let L be the set obtained by adding two elements $m \neq M$ to an infinite anti-chain[3] X. We extend the partial order from X to L as follows. For every $x \in X$, we require that $m \leq x \leq M$. Then L is a complete lattice. In fact, for every nonempty subset S of L, one has $\inf_L(S) = m$ and $\sup_L(S) = M$. The interval topology of L is the cofinite topology. Fix $x_0 \in X$. Then $L \setminus \{x_0\}$ is a subcomplete sublattice of L. It is compact but not closed in the interval topology of L.

The proof of Fact 1.1 also relies on Zhou's generalization (Fact 2.9) of Tarski's fixed point theorem [Tar55, Theorem 1]. For a self correspondence $Y : S \twoheadrightarrow S$ on a set S, let $\mathrm{Fix}(S) = \{x \in S : x \in Y(x)\}$.

Fact 2.9 (Tarski-Zhou, [Zho94, Theorem 1])**.** *Let S be a complete lattice. Let $Y : S \twoheadrightarrow S$ be an increasing correspondence, such that for every $x \in S$, $Y(x)$ is a nonempty* subcomplete *sublattice of S. Then* $\mathrm{Fix}(Y)$ *is a nonempty complete lattice.*

We shall prove two theorems announced without proof by Topkis. This requires Lemma 2.10.

Lemma 2.10.

(1) *Let P be a poset and Q be a subcomplete sublattice of P. Let τ_P (resp. τ_Q) be the interval topology of P (resp. Q). Then $\tau_Q = \tau_P|_Q$.*
(2) *Let $\{P_i\}_{i \in I}$ be a family of posets, and let τ_i be the interval topology of P_i. Denote the product topology of the topologies τ_i on $P := \prod_i P_i$ by τ. For each $i \in I$, let $L_i \subset P_i$ be a subcomplete sublattice. Then the restriction of τ to $\prod_i L_i$ is the interval topology of $\prod_i L_i$.*

Proof.

(1) For every $a \in Q$, the interval $[a, +\infty)_Q = [a, +\infty)_P \cap Q$ is closed in $\tau_P|_Q$. Similarly, $(-\infty, a]_Q$ is closed in $\tau_P|_Q$. Therefore, the topology τ_Q generated by these intervals is weaker than $\tau_P|_Q$.

[3]We recall that an anti-chain is a poset where any two distinct elements are incomparable.

For every $y \in P$, we show that $[y, +\infty)_P \cap Q$ is closed in τ_Q.

If this set is empty then it is trivially closed. If this set is nonempty, then as Q is subcomplete in P, we know that $z := \inf_P([y, +\infty)_P \cap Q)$ exists in Q. For any $x \in [y, +\infty)_P \cap Q$, we have $x \geq y$ and $x \geq z$. So $z \geq y$. Therefore, $[y, +\infty)_P \cap Q = [z, +\infty)_Q$ is closed in τ_Q.

Similarly, we can show that $(-\infty, y]_P \cap Q$ is closed in τ_Q. Thus, $\tau_Q \supset \tau_P|_Q$.

(2) As a property of product topology, the product of the topologies $\{\tau_i|_{L_i}\}_{i \in I}$ on $\prod_{i \in I} L_i$ is exactly the restriction $\tau|_{\prod_i L_i}$ of τ. For every $i \in I$, by Point (1), the interval topology σ_i of L_i equals $\tau_i|_{L_i}$. Moreover, the lattice L_i is complete hence bounded. By [AF66, Thm. 3], the interval topology of $\prod_i L_i$ is the product topology $\prod_i \sigma_i$.

□

Theorems 2.11 and 2.13 are stated in [Top98, p.31], whose proofs are referred to the unpublished work [Top]. They have applications to supermodular games. Theorem 2.11 appears without proof in [MS94, Footnote 3]. For the convenience of the reader, we include proofs.

Theorem 2.11. *Let L be a poset, and let $S \subset L$ be a sublattice. Then S is a subcomplete sublattice if and only if S is compact in the interval topology of L.*

Proof. First, assume that S is a subcomplete sublattice. Then S is a complete lattice. By the Frink-Birkhoff theorem ([Bir40, Thm. 20, p.250]), S is compact in its own interval topology. By Lemma 2.10 (1), S is compact in the interval topology of L.

Conversely, assume that S is compact in the interval topology of L. Then S is compact in its own interval topology. By the Frink-Birkhoff theorem, the lattice S is complete. In particular, $\max(S)$ exists.

For every nonempty subset A of S, we show that $\sup_L(A)$ exists and is in S. Let $B = \{x \in L : x \geq a, \forall a \in A\}$. Then $\max(S) \in B$. In particular, B is nonempty.

Consider a nonempty family of subsets $\{S \cap [a, b]_L\}_{a \in A, b \in B}$ of S. For every $a \in A$, every $b \in B$, the interval $[a, b]_L$ is closed in L, so $S \cap [a, b]_L$ is closed in S in the interval topology of L. Consider any nonempty finite subfamily: $\{S \cap [a_i, b_i]_L\}_{i=1}^m$, where $m \geq 1$ is an integer, $a_i \in A$, and $b_i \in B$. As S is a sublattice of L,

$$c := \sup_L\{a_i : 1 \leq i \leq m\} = \sup_S\{a_i : 1 \leq i \leq m\}$$

exists in S. By definition, $c \geq a_i$ for all $1 \leq i \leq m$. For every $1 \leq i \leq m$, as $b_i \in B$ we have $b_i \geq a_j$ for all $1 \leq j \leq m$, so $b_i \geq c$. Therefore, c is in the intersection of this finite subfamily.

By the compactness of S, the intersection of the whole family contains an element z. For every $a \in A$, every $b \in B$, one has $z \in S \cap [a, b]_L$, so $a \leq z$. Therefore, $z \in B$ is the least element of B. Thus, $\sup_L(A) = \min(B) = z$ exists and is in S.

Similarly, $\inf_L(A)$ exists and is in S. Therefore, the sublattice S is subcomplete. □

Remark 2.12. When $S = L$, then Theorem 2.11 reduces to the Frink-Birkhoff Theorem ([Bir40, Thm. 20, p.250]).

Theorem 2.13. *Let $\{P_i\}_{i \in N}$ be a nonempty family of posets, and let S be a nonempty sublattice of $P := \prod_i P_i$. Then S is a subcomplete sublattice of P if and only if S is compact in the product topology τ of the individual interval topology τ_i of each P_i.*

Proof. First, assume that $S \subset P$ is a subcomplete sublattice. For each $i \in N$, let L_i the projection of S on P_i. Then L_i is a subcomplete sublattice of P_i.

Also, S is a subcomplete sublattice of $\prod_i L_i$. By Theorem 2.11, S is compact in the interval topology σ of $\prod_i L_i$. By Lemma 2.10 (2), σ is the restriction of τ. Therefore, S is compact in the topology τ.

Conversely, assume that S is compact in τ. The interval topology of P is weaker than τ, so S is also compact in the interval topology of P. By Theorem 2.11, S is a subcomplete sublattice of P. □

Remark 2.14. In Theorem 2.13, if N is a singleton, then we recover Theorem 2.11. If $P_i = \mathbb{R}^1$ for all $i \in N$, then we recover [Top98, Theorem 2.3.1].

3. Supermodular game

In Section 3, standard definitions and basic facts related to supermodular games are recalled.

Definition 3.1. A noncooperative game $G = (N, \{S_i\}_{i \in N}, S, \{f_i\}_{i \in N})$ is the following data:

- a nonempty set of players N;
- for each $i \in N$, a nonempty set S_i of the strategies of player i;
- a nonempty subset $S \subset \prod_{i \in N} S_i$ of feasible joint strategies satisfying that each projection $S \to S_i$ is surjective;
- for each player $i \in N$, a payoff function $f_i : S \to \mathbb{R}$.

Remark 3.2. Definition 3.1 is slightly more general than that in [Top98, p.176]. According to [Top98, p.178], in real applications to game theory, a profile of strategies $x \in \prod_i S_i$ is feasible for each player $i \in N$ but may not be allowed as a joint strategy. So, Definition 3.1 is not limited to the case of $S = \prod_i S_i$.

For a noncooperative game $G = (N, \{S_i\}_{i\in N}, S, \{f_i\}_{i\in N})$, every joint strategy $x \in S$ and every player $i \in N$, let $x_{-i} \in \prod_{j\neq i} S_j$ denote the strategies of all players in N except player i for x. Put

$$S_i(x_{-i}) := \{y_i \in S_i : (y_i, x_{-i}) \in S\}.$$

It is nonempty, since $x_i \in S_i(x_{-i})$. For every player $i \in N$, let S_{-i} be the image of S under the projection $S \to \prod_{j\neq i} S_j$.

For $x \in S$, put

$$S(x) = (\prod_{i\in N} S_i(x_{-i})) \cap S.$$

Since $x \in S(x)$, $S(x)$ is nonempty. Put

$$\Delta := \{(y, x) \in S \times S : y \in S(x)\}.$$

Then Δ is a sublattice of $S \times S$.

Definition 3.3. [Top98, p. 178] A feasible joint strategy $x \in S$ is an equilibrium if $f_i(y_i, x_{-i}) \leq f_i(x)$ for every $i \in N$ and every $y_i \in S_i(x_{-i})$. The subset of S of all equilibria is denoted by E.

Set

$$F_i := \{x \in S : f_i(y_i, x_{-i}) \leq f_i(x), \quad \forall y_i \in S_i(x_{-i})\}. \tag{1}$$

Then

$$E = \cap_{i\in N} F_i. \tag{2}$$

Definition 3.4. [Top98, pp.177-178] For each player $i \in N$, the (individual) best response correspondence $Y_i : S \twoheadrightarrow S_i$ is defined by

$$Y_i(x) = \text{argmax}_{y_i\in S_i(x_{-i})} f_i(y_i, x_{-i}).$$

(Since $Y_i(x)$ depends only on x_{-i}, by abuse of notations, we sometimes write $Y_i(x_{-i})$ to emphasis this point.) More generally, when I is a *finite* nonempty subset of N, the best partial response correspondence $Y_I : S \twoheadrightarrow S$ relative to I is defined as

$$Y_I(x) = \arg\max_{y\in S(x)} g_I(y, x),$$

where for every $x \in S$, the function $g_I(\cdot, x) : S(x) \to \mathbb{R}$ is defined[4] by

$$g_I(y, x) = \sum_{i \in I} f_i(y_i, x_{-i}).$$

When N is finite, the best joint response correspondence is $Y = Y_N$.

The correspondence Y_I shall play a role in the case of infinitely many players. When $I = \{i\}$ is a singleton, one has

$$Y_I(x) = (Y_i(x) \times \prod_{j \neq i} S_j) \cap S(x).$$

The classical best response $R : S \twoheadrightarrow S$ is defined as $R(x) = (\prod_{i \in N} Y_i(x_{-i})) \cap S$. Here is a comparison of the two.

Lemma 3.5. *For every $x \in S$, every finite nonempty subset I of N, one has $R(x) \subset Y_I(x)$.*

Proof. For every $y \in R(x)$, $y' \in S(x)$, and $i \in I$, one has $y_i \in Y_i(x_{-i})$, thus $f_i(y_i, x_{-i}) \geq f_i(y'_i, x_{-i})$. Taking sum over $i \in I$, we get $g_I(y, x) \geq g_I(y', x)$, so $y \in Y_I(x)$. □

The correspondences Y and R are used to convert the problem concerning equilibria into a fixed point problem.

Fact 3.6. *We have $E = \mathrm{Fix}(R)$.*

Lemma 3.7. *For every finite nonempty subset I of N, $\mathrm{Fix}(Y_I) = \cap_{i \in I} F_i$. In particular, if N is finite, then $\mathrm{Fix}(Y_N) = E$.*

Proof. If $x \in S \setminus \cap_{i \in I} F_i$, then there is $i_0 \in I$ with $x \notin F_{i_0}$. In particular, there is $y_{i_0} \in S_{i_0}(x_{-i_0})$ with $f_{i_0}(x) < f_{i_0}(x')$, where $x'_{i_0} = y_{i_0}$ and $x'_i = x_i$ for all $i \neq i_0$. Consequently, we have $x' \in S(x)$. Now, for every $i \neq i_0$, $f_i(x'_i, x_{-i}) = f_i(x)$, so $g_I(x', x) > g_I(x, x)$, which implies

$$x \notin \mathrm{argmax}_{y \in S(x)} g_I(y, x) = Y_I(x),$$

and finally, $x \notin \mathrm{Fix}(Y_I)$.

Conversely, if $x \in \cap_{i \in I} F_i$, then for every $y \in S(x)$, every $i \in I$, we have $y_i \in S_i(x_{-i})$. So $f_i(y_i, x_{-i}) \leq f_i(x)$. Summing over $i \in I$, we get $g_I(y, x) \leq g_I(x, x)$. Therefore,

$$x \in \mathrm{argmax}_{y \in S(x)} g_I(y, x) = Y_I(x),$$

i.e., $x \in \mathrm{Fix}(Y_I)$.

The second part follows from the first one and (2). □

[4]Such a function is introduced in [NI55, (1), p.808] and [Ros65, Theorem 1] uses this construction to study the equilibria for games in non-product spaces.

Definition 3.8. A noncooperative game $G = (N, \{S_i\}_{i\in N}, S, \{f_i\}_{i\in N})$ is called supermodular if

- each S_i is a lattice and S is a sublattice of $\prod_i S_i$;
- for every $i \in N$, every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i(x_{-i}) \to \mathbb{R}$ is supermodular;
- for every $i \in N$, the function f_i has increasing difference relative to the subset S of $S_i \times (\prod_{j\neq i} S_j)$ in the sense of Definition 2.3.

Remark 3.9. Definition 3.8 extends the definition in [Top98, p.178] (which requires further that each S_i is a sublattice of some Euclidean space), and the definition in [Zho94, p.298] (which requires that $S = \prod_i S_i$ is in the product form).

By [Top98, Lemma 2.2.3, p.17], for every $i \in N$ and every $x \in S$, $S_i(x_{-i})$ is a sublattice of S_i. So we are allowed to talk about the supermodularity of the function $f_i(\cdot, x_{-i}) : S_i(x_{-i}) \to \mathbb{R}$. Moreover, for every $x \in S$, $\prod_i S_i(x_{-i})$ is a sublattice of $\prod_i S_i$. Therefore, $S(x)$ is a sublattice of $\prod_{i\in N} S_i$ and of S.

Lemma 3.10. *For a supermodular game as above,*

(1) *for every $i \in N$, the correspondence*

$$S \twoheadrightarrow S_i, \quad x \mapsto S_i(x_{-i})$$

is increasing;

(2) *the correspondence*

$$S \twoheadrightarrow S, \quad x \mapsto S(x)$$

is increasing.

Proof. For every $x \leq x'$ in S,

(1) for every $y_i \in S_i(x_{-i})$ and $y'_i \in S_i(x'_{-i})$, both (y_i, x_{-i}) and (y'_i, x'_{-i}) belong to S. Since S is a sublattice of $\prod_{i\in N} S_i$, we have

$$(y_i, x_{-i}) \wedge (y'_i, x'_{-i}) = (y_i \wedge_i y'_i, x_{-i}) \in S.$$

Thus, $y_i \wedge_i y'_i \in S_i(x_{-i})$. Similarly, $y_i \vee y'_i \in S_i(x'_{-i})$.

(2) For every $y \in S(x)$ and every $y' \in S(x')$, since S is a sublattice of $\prod_{i\in N} S_i$, we have $y \wedge y' \in S$. By Point (1), for every $i \in N$, $(y \wedge y')_i = y_i \wedge y'_i \in S_i(x_{-i})$, so $y \wedge y' \in S(x)$. Similarly, $y \vee y' \in S(x')$.

□

4. Main results

As recalled in Section 2, Zhou's proof of Fact 1.1 rests on Statement 2.7 which is false in general. In this section, we correct Zhou's proof and generalize Fact 1.1 in Theorem 4.2.

In Section 4, fix a supermodular game $G = (N, \{S_i\}_{i\in N}, S, \{f_i\}_{i\in N})$ and for every $i \in N$, consider a topology τ_i on S_i. Unless otherwise specified, we use the product topology on $\prod_i S_i$.

In the whole section, **we assume Assumption 4.1**.

Assumption 4.1.

- For every $i \in N$, the topology τ_i is finer than the interval topology of S_i;
- The sublattice $S \subset \prod_i S_i$ is closed compact in this topology;
- for every $i \in N$, every $x \in S$, the function $f_i(\cdot, x_{-i}) : S_i(x_{-i}) \to \mathbb{R}$ is upper semicontinuous.

We now state the main theorem of this note.

Theorem 4.2. *If I is a finite nonempty subset of N, then* $\mathrm{Fix}(Y_I)$ *is a nonempty complete lattice. In particular, if moreover N is finite, then E is a nonempty complete lattice.*

First, we prove several lemmata that we use in the proof of Theorem 4.2.

Lemma 4.3. *For every $j \in N$,*

(1) *S_j is a τ_j-compact complete lattice;*
(2) *for every $x \in S$, $S_j(x_{-j})$ is compact closed in S_j. The subset $S(x)$ is closed compact in $\prod_i S_i$.*

Proof.

(1) Since the projection $S \to S_j$ is continuous surjective, S_j is τ_j-compact. By the assumption on τ_j, S_j is compact in its interval topology. The completeness of S_j follows from the Frink-Birkhoff theorem.
(2) Consider the continuous injection

$$S_j \to \prod S_i, \quad y \mapsto (y, x_{-j}).$$

The preimage $S_j(x_{-j})$ of S is closed in S_j, thence also compact. Therefore, $\prod_i S_i(x_{-i})$ is compact by Tychonoff's theorem. Since S is closed in $\prod_i S_i$, $S(x)$ is closed in $\prod_i S_i(x_{-i})$. As $\prod_i S_i(x_{-i})$ is closed in $\prod_i S_i$, the subset $S(x)$ is closed in $\prod_i S_i$.

□

Lemma 4.4. *The sublattice S is subcomplete in $\prod_i S_i$.*

Proof. By assumption, S is compact in the product of individual interval product of each S_i. We conclude by Theorem 2.13. □

Lemma 4.5. *For every $i \in N$, every $x \in S$, the set $Y_i(x_{-i})$ is a nonempty closed compact subset of S_i.*

Proof. By Lemma 4.3, $S_i(x_{-i})$ is a nonempty compact space. By assumption, the function $f_i(\cdot, x_{-i}) : S_i(x_{-i}) \to \mathbb{R}$ is upper semicontinuous. Then

$$Y_i(x) = \text{argmax}_{y_i \in S_i(x_{-i})} f_i(y_i, x_{-i})$$

is a nonempty closed compact subset of $S_i(x_{-i})$. □

Lemma 4.6. *For every $i \in N$, the correspondence $Y_i : S \twoheadrightarrow S_i$ is increasing. In particular, for every $x \in S$, $Y_i(x)$ is a sublattice of S_i.*

Proof. For every $t \in S$, the function $f_i(\cdot, t_{-i}) : S_i \to \mathbb{R}$ is supermodular. Let $B = \{(x_i, t) \in S_i \times S | (x_i, t_{-i}) \in S\}$. The function

$$B \to \mathbb{R}, \quad (x, t) \mapsto f_i(x_i, t_{-i})$$

has increasing difference relative to $B \subset S_i \times S$. We conclude by Lemma 4.5 and Topkis' monotonicity theorem [Top98, Theorem 2.8.1]. □

Lemma 4.7. *For every finite nonempty subset I of N:*

(1) *The set $Y_I(x)$ is a nonempty closed* compact *subset of $\prod_i S_i$ for every $x \in S$.*
(2) *The correspondence $Y_I : S \twoheadrightarrow S$ is increasing. In particular, for every $x \in S$, $Y_I(x)$ is a sublattice of S and of $\prod_i S_i$.*
(3) *For every $x \in S$, the sublattice $Y_I(x)$ is* subcomplete *in S. In particular, both* $\max Y_I(x)$ *and* $\min Y_I(x)$ *exist.*

Proof.

(1) By assumption, for every $i \in N$, the function $f_i(\cdot, x_{-i}) : S(x) \to \mathbb{R}$ is upper semicontinuous. As the sum of finitely many upper semicontinuous functions, the function $g_I(\cdot, x) : S(x) \to \mathbb{R}$ is upper semicontinuous. By Lemma 4.3, $S(x)$ is nonempty compact closed subset of $\prod_i S_i$. Then $Y_I(x)$ is a nonempty closed compact subset of $S(x)$.
(2) The correspondence

$$S \twoheadrightarrow S, \quad x \mapsto S(x)$$

is increasing by Lemma 3.10 (2). For every $i \in N$, every $x \in S$, the function

$$S(x) \to \mathbb{R}, \quad y \mapsto f_i(y_i, x_{-i})$$

is supermodular by Definition 3.8. As the sum of finitely many supermodular functions, the function $g_I(\cdot, x) : S(x) \to \mathbb{R}$ is also supermodular. By Definition 3.8, the function

$$\Delta \to \mathbb{R}, \quad (y, x) \mapsto f_i(y_i, x_{-i})$$

has increasing difference relative to $\Delta \subset S_i \times (\prod_{j \neq i} S_j)$. As the sum of finitely many such functions, $g_I : \Delta \to \mathbb{R}$ also has increasing difference relative to $\Delta \subset S_i \times (\prod_{j \neq i} S_j)$. By Point (1) and [Top98, Theorem 2.8.1, p.76], the correspondence Y_I is increasing.

(3) By Point (1) and Theorem 2.13, $Y(x)$ is a subcomplete sublattice of $\prod_i S_i$, hence also a subcomplete sublattice of S.

□

Now, we prove the main theorem.

Proof of Theorem 4.2. We apply Fact 2.9. For this, we check the conditions in order. By Lemma 4.4, the lattice S is complete. By Lemma 4.7, the correspondence Y_I is increasing and for every $x \in S$, the set $Y_I(x)$ is a nonempty subcomplete sublattice of S. Now by Fact 2.9, $\mathrm{Fix}(Y_I)$ is a nonempty complete lattice. The second statement follows by Lemma 3.7. □

Remark 4.8. In Theorem 4.2, when each (S_i, τ_i) is a sublattice of some Euclidean space, we recover Fact 1.2. In another direction, if we take S to be the full $\prod_i S_i$, then we obtain Fact 1.1.

A noncooperative game is called *finite* if the player set and the strategy set of every player are finite.

Corollary 4.9. *For every finite supermodular game, the set of Nash equilibria is a nonempty lattice.*

Proof. For every $i \in N$, let τ_i be the discrete topology of S_i. Then Assumption 4.1 is satisfied. We conclude by Theorem 4.2. □

Remark 4.10. Although Corollary 4.9 seems fundamental, as far as we know it is not explicitly stated in the literature. A special case (with every S_i is a chain and $S = \prod_i S_i$, as remarked in [TY02, footnote on p.2]) is in [FM01, Theorem 3].

Theorems 4.11 and 4.12 are two results in the case of infinitely many players, but with some extra restrictions.

Theorem 4.11. *If moreover $S = \prod_i S_i$, then E is a nonempty complete lattice.*

Proof. By Lemma 4.5, for every $i \in N$, every $x \in S$, the set $Y_i(x_{-i})$ is nonempty and closed in S_i. Then $R(x) = \prod_i Y_i(x_{-i})$ is nonempty and closed in $S = \prod_i S_i$. As S is compact, $R(x)$ is compact in the product topology, and hence compact in the interval topology of S. By Lemma 4.4, the lattice S is complete. On account of Lemma 4.6, for every $i \in N$, the correspondence $Y_i : S \twoheadrightarrow S_i$ is increasing. Therefore the correspondence R is increasing. In particular, for every $x \in S$, $R(x)$ is a sublattice of S. By Theorem 2.11, $R(x)$ is a subcomplete sublattice of S. The proof is completed by Facts 2.9 and 3.6. □

If the game is not in the product form, we have only an existence result.

Theorem 4.12. *Assume further that for every $i \in N$, the function $f_i : S \to \mathbb{R}$ is continuous and the correspondence*

$$\phi_i : S_{-i} \twoheadrightarrow S_i, \quad x_{-i} \mapsto S_i(x_{-i})$$

is lower semicontinuous. Then E is nonempty.

Proof. For every $i \in N$, we show that the F_i defined in (1) is closed in S.

In fact, take any $s \in S \setminus F_i$, there is $y_i \in S_i(s_{-i})$ such that $f_i(s) < f_i(y_i, s_{-i})$. Take $a \in (f_i(s), f_i(y_i, s_{-i})) \subset \mathbb{R}$. By the continuity of $f_i : S \to \mathbb{R}$, the set $W := \{x \in S : f_i(x) > a\}$ is an open neighborhood of $(y_i, s_{-i}) \in S$. By the definition of product topology on $S_i \times S_{-i}$, there exist open subsets $U \subset S_i$ and $V \subset S_{-i}$ with $(y_i, s_{-i}) \in (U \times V) \cap S \subset W$. Then $y_i \in U$ and $s_{-i} \in V$.

Set

$$O := \{x_{-i} \in S_{-i} : S_i(x_{-i}) \cap U \neq \emptyset\}.$$

By the lower semicontinuity of ϕ_i, O is open in S_{-i}. As $y_i \in U \cap S_i(s_{-i})$, one has $s_{-i} \in O$. By the continuity of f_i on S again, the subset

$$A = (S_i \times (V \cap O)) \cap \{x \in S : f_i(x) < a\}$$

is open in S and contains s. For every $x \in A$, one has $f_i(x) < a$ and $x_{-i} \in O \cap V$, so there is $y_i' \in S_i(x_{-i}) \cap U$. Then

$$(y_i', x_{-i}) \in (U \times V) \cap S \subset W,$$

so $f_i(y_i', x_{-i}) > a > f_i(x)$. Thus, $x \notin F_i$ for all $x \in A$, or equivalently $A \subset S \setminus F_i$. This shows the closedness of F_i.

For every finite nonempty subset I of N, by Lemma 3.7, $\cap_{i \in I} F_i = \mathrm{Fix}(Y_I)$. It is nonempty by [Top98, Lemma 2.2.3, p.17] and Theorem 4.2. By the compactness of S, the full intersection $E = \cap_{i \in N} F_i$ is nonempty. □

Remark 4.13. A closely related result is [Top98, Theorem 4.3.2, p.189], which concerns the algorithmic aspect with N being finite, while we study the existence of equilibria with N being infinite.

Statements and Declarations

Conflict of interests. The author declares that she has no conflict of interests.

Ethical approval. This article does not contain any studies with human participants or animals performed by the author.

Informed consent. For this type of study informed consent was not required.

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